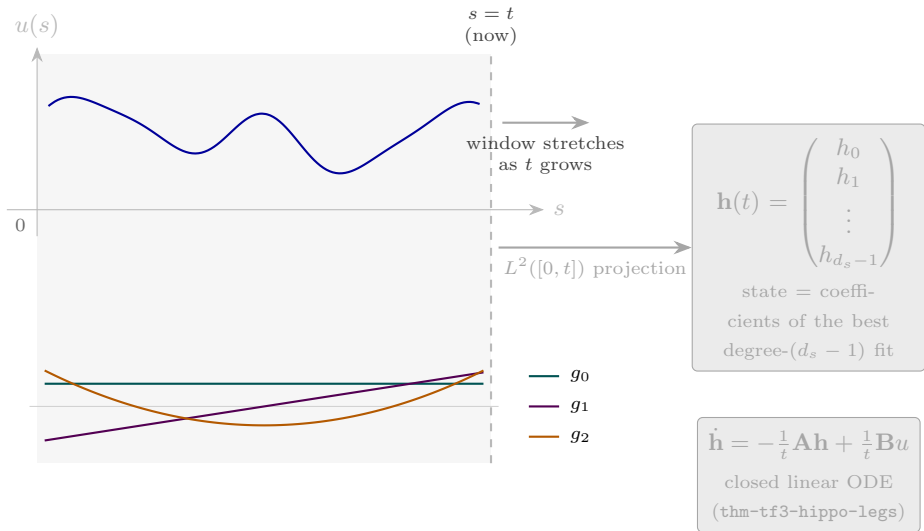




The state is not an ad-hoc memory: $\mathbf{h}(t)$ is the $L^2([0, t])$ -optimal coefficient vector of the best degree- $(d_s - 1)$ polynomial fit to the whole history $u|_{[0, t]}$, and maintaining it online is exactly this linear state-space recursion.